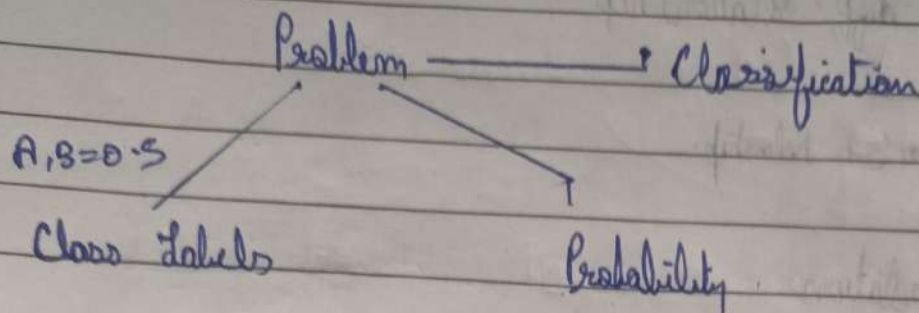


Metric in "Classification"



- (i) Confusion Matrix
- (ii) FPR (Type I error)
- (iii) FNR (Type II error)
- (iv) Recall (TPR, Sensitivity)
- (v) Precision (True prediction value)
- (vi) Accuracy
- (vii) F Beta
- (viii) Cohen Kappa
- (ix) ROC Curve, AUC Score
- (x) PR curve.

Confusion matrix

← Actual Value.

	1	0	
Predicted Value	1 True +ve	False +ve	→ Type I error ↓ False +ve Rate FPR ⇒ $\frac{FP}{FP + TN}$
	False -ve	True -ve	
	↓ Type II error False -ve Rate FNR		

Accuracy → $\frac{TP + TN}{TP + FP + TN + FN}$

day in a dataset

A $\rightarrow$ 900 , B $\rightarrow$ 100

$\rightarrow$ using Accuracy matrix on an imbalanced dataset.

- Recall $\rightarrow$ Out of all +ve values (TP, FN) how many (TPR) did our model predict correctly

$$\frac{TP}{TP + FN}$$

• Precision $\rightarrow$ Out of total predicted +ve results, how many (true posⁿ value) did we actually predict right

$$\frac{TP}{TP + FP}$$

day in confusion matrix,

day Cancer - detection

	0	1
0	TP	FP
1	FN	TN

in these cases
we use Recall

$\downarrow \downarrow \rightarrow$ since in 1 column of FN -
if our model predicted
falsely Negative (Person doesn't have Cancer)
It would be a disaster

- If FN imp $\rightarrow$ Use Recall
- If FP imp $\rightarrow$ Use Precision

$\rightarrow F \cdot \beta$

F-Beta score =

$$(1 + \beta^2) \cdot \frac{\text{Precision} \times \text{Recall}}{\beta^2 \times \text{Precision} + \text{Recall}}$$

$\beta = 1$ F_1 score

$\beta = 0.5$ F0.5 Score

Q1 $\beta = 1$

$$2 \cdot \frac{P \times R}{1 \times P + R} \rightarrow \frac{2PR}{P+R} \text{ a Harmonic Mean.}$$

Q1 FP & FN are imp $\beta = 1$

FP is imp $\beta = -$ (P val dec)
FN is imp $\beta = ++$ (P val inc)

Q1 FN impbal is great
then $\downarrow$
 $0 < \beta < 10$